The Rising Sun: Shopping List

For the first prototype, the light only. Written 2026-10-18 for Dr. O.

No prices are listed because I cannot check them from here. Nothing below has been tried on real parts. If a listing does not match what is written, do not buy it, and ask first.

Buy for the first prototype

- Controller board: you already have the ESP32-S3 N16R8 from the Holopod build. Use one of those. Nothing to buy.
- LED strip: search "WS2812B LED strip 5V". It must say 5V, WS2812B (or WS2812), and "individually addressable". Buy 1 meter with 60 LEDs per meter. Black or white backing both work. Do NOT buy 12V strips, or strips that say "RGB" with only 4 wires and no data line.
- Power supply: a 5V supply rated 4 amps or more, with a screw terminal or a barrel plug and a matching adapter. 60 LEDs can draw up to 3.6 amps at full white. Do NOT use a phone charger.
- Resistor: 330 ohm, any small assortment kit has it.
- Capacitor: 1000 microfarad, 6.3V or higher. Also in most kits.
- Jumper wires: male to female and male to male, a small assortment.
- Breadboard or a few screw terminal blocks, so nothing has to be soldered for the first try.
- USB cable: it must carry data, not just power. If the board does not show up on the computer, a charge-only cable is the usual reason.

Buy only if the first light flickers

- A level shifter, 74AHCT125 (a small board, sold as "logic level shifter"). The board sends a 3.3V signal and some strips want 5V.

Buy for sound: the speakers and the microphone

Buy these with the first order. They are wired after David has seen the light working, and the program that plays her voice and music is not written yet. The parts below are from memory and have not been checked against a build, so look at each listing before buying.

- Speakers: the same ones as Astra-9. Use "Buy it again" on your Amazon order of September 15, 2026: "2pcs 3 W 8 Ohm Mini Speaker JST-PH2.0mm 2 pin interface with Mini PAM8403 DC 5V Audio Amplifier Digital Power Module 3W+3W". They are 3 watt, 8 ohm, so they work with the amplifiers below. The amplifier modules in that order are a spare, because

they take analog sound and the sun board has no analog sound output. Order one extra set if you want a spare pair of speakers.
- Amplifiers: two I2S amplifier boards, MAX98357A, one for each speaker. Each takes digital sound from the board and drives one speaker, so no separate sound card is needed. Check that the listing says I2S and 3 watts. They drive 8 ohm speakers a little more quietly than 4 ohm ones, which is fine for a bedroom.
- Microphone: one I2S microphone board, INMP441, so she can hear you. (A second one gives better pickup if the acrylic muffles sound.)
- Wire and terminals: the speakers come with small JST plugs, and the amplifier boards use screw terminals or solder pads. The simplest fix is to cut the plug off each speaker lead and put the bare wire in a screw terminal, the same green blocks that held up best on Astra-9. Add a few more jumper wires. Support every wire so nothing hangs from a joint.
- Spare board: a second ESP32-S3 board, if the Holopod build does not have one free. The sun board and the Holopod board cannot be the same board.

Important about the board: the ESP32-S3 has Wi-Fi and low energy Bluetooth but not the regular Bluetooth that phone speakers use. The sun cannot be a Bluetooth speaker. Music will come from a service such as Spotify, through Wi-Fi.

Where the speakers go: behind or beside the acrylic, facing out, with some air around them. They are sound parts, so keep them away from the LED strip power wires to avoid hum. The final wiring list will come with the sound program.

Do not buy yet

- The full length of strip for the acrylic. How many LEDs it needs depends on the acrylic shape, which is not decided. Buy it after David has seen the prototype and you have picked the layout.
- The bigger power supply. It is sized from the final LED count: about 60 milliamps per LED at full white, and a safe supply has some room above that.
- The sound program. It is not written yet. The parts for it are listed above.

Before David starts

- Lay the parts out and tick them off against this list.
- Give him the First Prototype sheet. It lists every step.
- He does not need any account, login or key for the first prototype.
- Ask him to send you a photo of the strip lit. That is the only report you need today.